Home4U

CSC 648 / CSC 848

Milestone 4 Version 1

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Section: 03

Team Number: 4
Team Name: Vibecoding for Internship
Project Name: Home4U
Team Lead: Caleb Ponce
Team Lead Email: cponce8@sfsu.edu
Team Members and Roles
Caleb Ponce — Team Lead / System Architecture
Mason Lee — Data Modeling & Scoring Engine
Christopher Quach — Frontend Development
Tyler Morris — Backend & AI Integration
Dias Almat — Technical Writer

Revision History:

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1. Product Summary:

1.1 Product Name

Home4U

1.2 Final Priority 1 and Priority 2 Functional Requirements

Home4U is a web-based room planning and interior design recommendations platform that helps users evaluate design styles, organize room projects, and generate structured room-improvement recommendations based on room type, style direction, and budget context.
The team's final committed functional scope is divided into Priority 1 and Priority 2 requirements.

Priority 1 Functional Requirements

1. The system shall allow users to create an account and log into a protected session.
2. The system shall allow authenticated users to create, view, update, and manage room projects.
3. The system shall allow authenticated users to create, view, update, and manage room projects.
4. The system shall allow users to launch a workspace from a selected style context.
5. The system shall allow users to upload a room image to a project.
6. The system shall allow users to set room type, budget tier, lighting context, and style intensity for analysis.
7. The system shall analyze saved room inputs and generate style scores, matched design signals, and prioritized recommendations.
8. The system shall persist project analysis results so users can revisit a saved project later.

9. The system shall generate a structured shopping plan and recommendation list tied to the saved room project.
10. The system shall allow users to review project details and mark recommendations complete.

Priority 2 Functional Requirements

1. The system shall provide a richer recommendation review experience through a dedicated Project Details page
2. The system shall provide budget-aware recommendation estimates and shopping guidance.
3. The system shall provide matched design-tag explanations to improve transparency of recommendations.
4. The system shall preserve uploaded room photos within saved project views.
5. The system shall provide accessibility improvements across major user flows.
6. The system shall support stable use across common desktop browsers during beta testing.

1.3 Unique or Superior Features

Home4U distinguishes itself from general inspiration platforms by focusing on structured, explainable project guidance instead of only visual browsing. Rather than showing disconnected inspiration images, the system helps users move from style selection to a saved room project, a budget-aware recommendation set, and a shopping-oriented action plan.
The system's strongest distinguishing features are:

- Explainable room-style scoring based on saved room signals, selected style attributes, and matched design tags.
- Budget-aware recommendation generation that helps users prioritize which changes to make first.
- Persisted room projects that allow users to revisit previous analyses instead of restarting each session.
- A project workflow that connects style discovery, room analysis, recommendations, and shopping guidance in one system.
- A structured recommendation process that is deterministic and easier to justify during testing and review.

1.4 Deployment URL

The deployed application URL for the current beta prototype is:

http://18.225.42.247/

2. Usability Test Plan:

2.1 Test Objective

The purpose of the usability test plan is to evaluate whether users can successfully understand and complete the major Home4U workflow without confusion, excessive assistance, or major navigation issues. Since Milestone 4 focuses on beta-level readiness, this test plan is intended to measure how clearly the system supports real users as they move through room planning, style review, workspace analysis, and project follow-up tasks.

The usability evaluation focuses on five major application functions, excluding login and registration as required by the milestone instructions. These functions were selected because they represent the main user journey and the most important feature set currently implemented in the deployed system.

2.2 Functions Selected for Testing

The following five major functions were selected for usability testing:

1. Browsing and selecting a design style from the Dashboard.
2. Launching the Workspace from a chosen style context.
3. Creating or updating a room project with room type and budget information.
4. Uploading a room image and generating a recommendation plan.
5. Viewing a saved project and reviewing recommendations in the Project Details page.

These tasks were selected because they reflect the central value of the product and exercise the most important user-facing flows in the current beta prototype.

2.3 Test Participants

Usability test participants must not be members of the development team. Participants should represent general users who are capable of browsing a web application and completing guided task flows, but who are not already familiar with the internal architecture or development details of Home4U.

Suggested participant profile:

- adults familiar with basic web navigation
- users with interest in room planning, apartment setup, or interior design
- users with no prior exposure to the Home4U internal workflow

Participant table:

Participant	Background	Technical Comfort	Prior Familiarity with Home4U
Participant 1	[Fill In]	[Low / Medium / High]	None
Participant 2	[Fill In]	[Low / Medium / High]	None
Participant 3	[Fill In]	[Low / Medium / High]	None

2.4 Test Environment and Setup

The usability test will be performed using the deployed Home4U beta prototype hosted on the team's AWS instance.

System Under Test: Home4U Beta Prototype
Deployment URL: http://18.225.42.247/
Devices Used: [Fill In]
Browsers Used: [Fill In]
Internet Connection: Stable broadband connection
Test Facilitator: [Fill In]
Test Location: [Fill In]

Each participant will be asked to complete a guided sequence of realistic tasks while the facilitator observes:

- completion success
- time spent
- visible hesitation points
- requests for clarification
- user comments and satisfaction feedback

2.5 Task Scenarios

Task 1: Browse and Select a Design Style

The participant is asked to explore the available styles on the Dashboard and choose a style they believe fits a target room direction.
Objective: Determine whether users can understand the style library and identify how to move from style browsing into the next step of the workflow.

Task 2: Launch the Workspace

The participant is asked to open the Workspace from the selected style.
Objective: Determine whether users understand how the selected style connects to the workspace flow and whether the transition is intuitive.

Task 3: Create or Update a Room Project

The participant is asked to choose a room type and set a project budget.
Objective: Determine whether users can identify the necessary inputs for beginning a room-planning workflow and whether the controls are understandable.

Task 4: Upload a Room Image and Generate a Plan

The participant is asked to upload a room image and generate a recommendation plan.
Objective: Determine whether users understand what inputs are needed for analysis and whether the generation flow feels clear and predictable.

Task 5: Review Saved Project Recommendations

The participant is asked to open a saved project and review the resulting recommendation and shopping-plan information.
Objective: Determine whether users can interpret the returned output and understand what actions they can take next.

2.6 Effectiveness Measurements

Effectiveness is measured by whether participants are able to complete tasks accurately and independently.
The following effectiveness criteria will be used:

- task completion rate
- number of failed tasks
- number of user errors
- number of times facilitator assistance was required
- number of task restarts or abandoned attempts

Suggested effectiveness table:

Task	Completed Without Help	Completed With Help	Not Completed	Notes
Task 1	[Fill In]	[Fill In]	[Fill In]	[Fill In]

Task 2	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 3	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 4	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 5	[Fill In]	[Fill In]	[Fill In]	[Fill In]

2.7 Efficiency Measurements

Efficiency is measured by the speed and smoothness with which participants can complete each task.
The following efficiency criteria will be used:

- time to complete each task
- number of clicks or major actions
- hesitation or confusion points
- unnecessary navigation steps
- repeated actions caused by unclear UI behavior

Suggested efficiency table:

Task	Avg. Time	Avg. Clicks / Actions	Hesitation Observed	Notes
Task 1	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 2	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 3	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 4	[Fill In]	[Fill In]	[Fill In]	[Fill In]
Task 5	[Fill In]	[Fill In]	[Fill In]	[Fill In]

2.8 User Satisfaction Survey

After task completion, each participant will complete a short satisfaction survey using a five-point Likert scale.
Scale:
1 = Strongly Disagree
2 = Disagree
3 = Neutral
4 = Agree
5 = Strongly Agree

Suggested survey questions:

Question	1	2	3	4	5
The system was easy to navigate.					
I understood what to do at each step.					
The labels and page content were clear.					
The recommendation workflow made sense to me.					
I felt confident using the product without assistance.					
The saved project and recommendation information was useful.					
I would be willing to use this product again.					

2.9 Observations and Results Summary

This subsection should summarize the actual outcomes after testing is completed.
Suggested summary points to fill in:

- which tasks had the highest completion rate
- which tasks caused the most hesitation
- whether users understood the relationship between style selection, workspace analysis, and saved project review
- whether recommendation outputs were understandable
- whether navigation labels matched user expectations
- what UI or wording changes should be made before Milestone 4 Version 2

Example structure:
The usability test results showed that participants were generally able to navigate the core Home4U workflow, especially when

browsing styles and launching the workspace. The most common hesitation points occurred during project setup and recommendation interpretation, where some users needed additional clarity about what inputs were required and what the generated outputs represented. Overall, the results suggest that the beta prototype supports the main workflow, but additional refinements to terminology, guidance, and result explanation would improve usability before the next milestone revision.

2.10 Planned Follow-Up Improvements

After usability testing, the team will review the observations and identify concrete improvements for Milestone 4 Version 2. These may include:

- clarifying labels and button text
- improving guidance in the workspace flow
- making recommendation outputs easier to interpret
- reducing confusion between project setup and final review
- improving consistency between wireframe expectations and the implemented interface

3. QA Test Plan

3.1 Test Objective

The objective of the QA test plan is to verify that Home4U satisfies key non-functional requirements expected of a beta prototype. While functional testing confirms that the system can perform required tasks, this QA plan focuses on how well the system performs under expected usage conditions, how safely it handles protected actions and invalid input, how reliably it responds to failures, and how consistently it behaves across supported browsers and environments.

This test plan selects five non-functional requirements from different categories and defines test cases for each. The selected categories are:

- Performance
- Security
- Reliability
- Usability / Accessibility
- Compatibility

3.2 Test Environment

Application Under Test: Home4U Beta Prototype
Deployment URL: http://18.225.42.247/
Backend Stack: FastAPI + SQLAlchemy + SQLite
Frontend Stack: React + Vite
Operating Systems Used: [Fill In]
Browsers Used: Chrome, Safari, Firefox [adjust if needed]
Devices Used: [Fill In]
Network Conditions: Standard residential broadband / campus Wi-Fi
Testers: [Fill In]

If needed, include both local validation and deployed-environment validation in the final version.

3.3 Non-Functional Requirement 1: Performance

Requirement
The system shall respond within an acceptable time during normal user interaction and shall remain usable while navigating core application flows.
Test Objective
To verify that the Dashboard, Workspace, Project Details, and recommendation flows remain responsive under normal beta-level usage.
HW and SW Setup

- Desktop or laptop computer
- Stable internet connection
- Browser: Chrome / Safari / Firefox
- Deployed Home4U beta application

Feature Under Test

- Dashboard page load
- Workspace generation flow
- Saved project loading
- Search responsiveness

Test Cases

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
P1	Open the deployed Dashboard after authentication.	Dashboard loads without crash and primary UI becomes usable within an acceptable time.	[Fill In]	[Fill In]

P2	Search for a style term such as "modern" or "scandinavian".	Search results appear without freezing the UI and can be interacted with normally.	[Fill In]	[Fill In]
P3	Launch Workspace, upload a room photo, and generate a plan.	The system completes analysis and displays results without timeout or unresponsive behavior.	[Fill In]	[Fill In]

Notes

Current local engineering validation shows that the frontend production build succeeds and major smoke tests complete successfully, but browser-observed timing results should still be recorded here from actual QA execution.

3.4 Non-Functional Requirement 2: Security

Requirement

The system shall protect authenticated user data and restrict unauthorized access to protected actions and resources.

Test Objective

To verify that protected routes require authentication, that unauthorized users cannot mutate protected data, and that sensitive authentication behavior is enforced correctly.

HW and SW Setup

- Desktop or laptop computer
- Browser or API client
- Deployed app and/or local development environment
- Test user account

Feature Under Test

- Protected routes
- JWT-based authentication
- Auth-protected data mutation
- Password handling behavior

Test Cases

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
S1	Attempt to access protected pages without being logged in.	User is redirected to the login page or denied access.	[Fill In]	[Fill In]
S2	Attempt to call a protected backend mutation endpoint without valid authentication.	Request is rejected with an authentication error.	[Fill In]	[Fill In]
S3	Attempt repeated invalid logins using the same credentials or IP.	The login endpoint eventually rate-limits the requests and returns a blocked response.	[Fill In]	[Fill In]

Notes

Current engineering verification confirms:

- protected routes redirect unauthenticated users
- the tag-creation endpoint now requires authentication
- auth rate-limit smoke tests pass locally

Include screenshots or API response evidence in the final document.

3.5 Non-Functional Requirement 3: Reliability

Requirement

The system shall handle invalid input, missing data, and internal failures gracefully without crashing the entire application.

Test Objective

To verify that Home4U remains stable when requests fail, when required project state is missing, or when unsupported input is provided.

HW and SW Setup

- Desktop or laptop computer
- Browser and/or API client
- Deployed app

- Test image files and malformed input where applicable

Feature Under Test

- Backend exception handling
- Upload validation
- Project analysis recovery
- API failure handling

Test Cases

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
R1	Upload an unsupported file type or invalid image to a project.	The request is rejected with a clear validation error and the app remains stable.	[Fill In]	[Fill In]
R2	Request project analysis for a project that does not exist or is not accessible to the current user.	The backend returns an appropriate error response without crashing.	[Fill In]	[Fill In]
R3	Simulate backend unavailability or API failure during frontend use.	The application shows a failure state or warning instead of crashing completely.	[Fill In]	[Fill In]

Notes

Current local code behavior includes:

- upload type and size validation
- explicit 404 and 400 responses for invalid project and recommendation conditions
- a frontend API status banner for backend reachability issues

3.6 Non-Functional Requirement 4: Usability / Accessibility

Requirement

The system shall provide understandable navigation, accessible controls, and usable interaction patterns for major user flows.

Test Objective

To verify that the system supports keyboard interaction, semantic labeling, understandable route flow, and accessible state communication across critical pages.

HW and SW Setup

- Desktop or laptop computer
- Keyboard-only interaction where applicable
- Browser: Chrome / Safari / Firefox
- Screen-reader spot checks if performed

Feature Under Test

- Dashboard interaction controls
- Workspace comparison controls
- Route transitions
- Form labeling and state communication

Test Cases

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
U1	Navigate to major pages and interact with buttons and forms using keyboard input.	Major controls are focusable and usable without mouse-only dependence.	[Fill In]	[Fill In]
U2	Use the Workspace comparison slider and observe accessible state text.	The control exposes a meaningful label and updates its state appropriately.	[Fill In]	[Fill In]
U3	Trigger protected-route navigation while unauthenticated.	The redirection behavior is predictable and understandable to the user.	[Fill In]	[Fill In]

Notes

Current local engineering validation confirms:

- frontend accessibility smoke tests pass
- Dashboard and Workspace accessibility checks pass in automated test coverage
- protected-route behavior works as expected in route smoke tests

3.7 Non-Functional Requirement 5: Compatibility

Requirement

The system shall behave consistently across supported browsers and standard desktop environments during normal beta use.

Test Objective

To verify that the deployed Home4U beta prototype is usable across multiple major browsers without layout breakage, blocking script errors, or major workflow regressions.

HW and SW Setup

- Desktop or laptop computer
- Browsers: Chrome, Safari, Firefox [adjust to actual browsers tested]
- Deployed Home4U application

Feature Under Test

- Dashboard
- Workspace
- Project Details
- About page
- Core authenticated navigation

Test Cases

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
C1	Open the Dashboard and browse styles in Chrome, Safari, and Firefox.	Layout renders correctly and interactive elements remain usable in each browser.	[Fill In]	[Fill In]
C2	Launch Workspace and complete the analysis flow in each tested browser.	Core workflow remains functional and visually stable.	[Fill In]	[Fill In]
C3	Open a saved project and review recommendations and shopping-plan details in each tested browser.	Project details render correctly and remain readable and interactive.	[Fill In]	[Fill In]

Compatibility Summary Table

Browser / Environment	Tested Areas	Result	Notes
Chrome	[Fill In]	[Fill In]	[Fill In]
Safari	[Fill In]	[Fill In]	[Fill In]
Firefox	[Fill In]	[Fill In]	[Fill In]

3.8 QA Results Summary

The QA test plan evaluates Home4U’s beta readiness across performance, security, reliability, usability/accessibility, and compatibility. Based on current engineering validation, the system already demonstrates stable frontend builds, passing smoke tests, protected-route enforcement, login rate limiting, and successful room-analysis workflows. However, the final Milestone 4 Version 1 document should include executed browser-based outcomes, screenshots, and any failed or partially passed cases discovered during team QA testing.
Any failed tests or partial results should be documented honestly and used to guide improvements for Milestone 4 Version 2.

4. AI Feature Testing and Ethics Review

Home4U uses AI-assisted features to support the interior design recommendation workflow. The AI helps analyze uploaded room images, suggest room tags, and support recommendation generation. However, the AI is not used as a final decision maker. Users remain responsible for reviewing, confirming, editing, or rejecting AI-generated suggestions before those suggestions are used in the recommendation workflow.

The purpose of this section is to document how Home4U validates AI behavior, handles AI limitations, protects users from misleading outputs, and ensures that AI-assisted recommendations remain transparent, fair, and user-controlled.

4.1 AI-Assisted or AI-Powered Features

- AI Room Image Analysis
- AI Tag Suggestions
- Resemblance Score Computation
- Budget Aware Recommendation Generation

4.2 Correctness and consistency of outputs

Area Tested	Expected Results	Outcome
Uploaded room image analysis	The system identifies visible room characteristics from the uploaded image.	

AI tag suggestions	Suggested tags describe room features such as furniture, colors, layout, and lighting.	
Confirmed RoomTags	User-reviewed tags are stored as confirmed RoomTags.	
Resemblance score	The system compares RoomTags with StyleTags and calculates a normalized score.	
Recommendation output	Recommendations are based on confirmed tags, selected style, resemblance score, and budget.	

4.3 Bias and fairness considerations

Area Tested	Expected Results	Outcome
Budget-aware recommendations	Recommendations stay within the user-defined budget when applicable.	
Multiple style comparison	The system supports comparing multiple selected styles.	
Style scoring	Each selected style is scored using the same weighted resemblance scoring process.	
User tag confirmation	Users can review and correct AI-suggested tags before scoring.	

Actionable recommendations	Recommendations are prioritized by impact-to-cost ratio.	

4.4 Input Validation and prompt constraints

Area Tested	Expected Results	Outcome
Image upload validation	The system validates the uploaded room image file type.	
Budget input validation	The system validates the budget amount entered by the user.	
User credential validation	The system validates user credentials before protected resources are accessed.	
Confirm Tags workflow	Users can edit or confirm suggested tags before they are stored.	
Protected recommendation workflow	Only authenticated users can access their own room projects and recommendations.	

4.5 Transparency of AI behavior to users

Area Tested	Expected Results	Outcome

AI tag suggestions	Users can see that tags are suggested by the system.	
Tag review step	Users review, edit, or confirm suggested tags before they are used.	
Resemblance score	Users can view the calculated similarity score.	
Recommendation explanation	Recommendations are connected to missing style tags, score, and budget.	
User control	Users choose the style, confirm tags, input budget, and decide whether to follow recommendations.	

4.6 Failure modes and safe degradation

Area Tested	Expected Results	Outcome
Uploaded room image analysis	The system rejects the file and does not continue with analysis.	
AI tag suggestion unavailable	Users can still manually enter or confirm room tags.	
No confirmed tags	The system should not compute a resemblance score until tags are available.	
Recommendation generation issue	The system displays an error instead of crashing.	
Unauthorized project access	The system blocks users from accessing projects that are not theirs.	

4.7 Confirmation That AI is an Assistant, not a decision maker

Area Tested	Expected Results	Outcome
Room image	User uploads the room image	
AI tag suggestions	User reviews, edits, or confirms suggested tags. furniture, colors, layout, and lighting.	
Tags	User selects the target style or styles.	
Style	User enters the budget constraint.	
Final decision	User decides whether to follow the generated recommendations.	

4.8 Ethical considerations and mitigation strategies

Ethical Consideration	Mitigation Strategy
Incorrect AI tag suggestions	Allow users to edit or confirm suggested tags before storing them.
Misleading recommendations	Base recommendations on confirmed tags, resemblance scores, and budget.
Budget limitations	Use budget-aware prioritization so recommendations consider the user's spending limit.
User data access	Enforce project ownership so users can access only their own projects.
Password security	Store passwords using hashing and salting.
Authentication security	Use JWT authentication for protected resources.

Uploaded image handling	Store uploaded images as room project data and link them to the correct user project.
AI overreach	Keep AI as a tag suggestion assistant; users confirm tags and make final decisions.

5. Localization Testing

5.1 Test Objective

The objective of localization testing is to evaluate whether Home4U can display user-facing content, formatting, and layout behavior in a way that remains understandable and stable across different locale assumptions. Although the current beta prototype does not implement a full multilingual interface or language-switching feature, localization testing is still necessary to determine how the application behaves when locale-sensitive data such as dates, currency, and text length vary.

This test plan focuses on four required localization areas:

- text behavior
- formatting behavior
- layout stability
- directionality readiness

The purpose of this section is to document the current localization readiness of the beta prototype, identify limitations, and define the risks that would need to be addressed in future versions if multilingual support were added.

5.2 Current Localization Scope

At the current beta stage, Home4U supports only a limited localization scope.

The current implementation includes:

- English-only interface text across the application
- browser-based locale formatting for some date values
- browser-based number formatting for some numeric and currency-related values
- no language selector
- no translation files
- no right-to-left (RTL) layout support
- no region-specific content customization

This means the current system should be evaluated as partially localization-aware in formatting, but not yet localized in the broader product sense.

5.3 Areas Under Test

The following localization areas were selected for review:

Text

To determine whether all visible labels, button text, messages, and content remain understandable and consistent under the current English-only implementation.

Formatting

To determine whether dates, numbers, and monetary values are displayed consistently and whether they behave reasonably when locale-sensitive browser behavior is involved.

Layout

To determine whether the interface remains visually stable if text becomes longer, shorter, or differently structured under future localization scenarios.

Directionality

To determine whether the current UI structure appears adaptable to right-to-left languages such as Arabic or Hebrew, even though RTL is not currently implemented.

5.4 Test Environment

Application Under Test: Home4U Beta Prototype
Deployment URL: http://18.225.42.247/
Browsers Used: Chrome, Safari, Firefox [adjust to actual browsers tested]
Devices Used: [Fill In]
Locale Conditions Reviewed: English (default), browser locale variation where applicable
Testers: [Fill In]

5.5 Localization Test Cases

5.5.1 Text Testing

Test Objective:
To verify that the current English interface uses clear, consistent wording and to identify whether future translation expansion would likely cause ambiguity.

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail

L1	Review the major pages including Dashboard, Workspace, Project Details, and About.	All major labels, actions, and section headings remain understandable and internally consistent in English.	[Fill In]	[Fill In]
L2	Review system notices, empty states, and API-related status messages.	User-facing messages remain readable and understandable without truncation or broken phrasing.	[Fill In]	[Fill In]
L3	Review recommendation and shopping-plan content for wording consistency.	Generated content remains understandable and uses stable terminology across pages.	[Fill In]	[Fill In]

Observed Risk:
Because the interface is currently authored only in English and does not use a translation framework, future localization would require manual extraction and replacement of hardcoded strings.

5.5.2 Formatting Testing

Test Objective:
To verify that date, number, and monetary information displays in a predictable way and to identify formatting assumptions that are not yet fully locale-safe.

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
L4	Review project creation dates on Dashboard and related project views.	Dates render correctly and remain readable under normal browser locale behavior.	[Fill In]	[Fill In]
L5	Review budget and recommendation cost values in Dashboard, Workspace, and Project Details.	Numeric values render clearly, with separators appropriate to the browser locale where supported.	[Fill In]	[Fill In]

L6	Compare date and number display across browsers with different locale settings if available.	Locale-sensitive formatting remains stable and does not break page structure.	[Fill In]	[Fill In]

Observed Risk:
Home4U currently mixes locale-aware number formatting with hardcoded U.S. dollar symbols such as $. This means formatting is only partially localized. Number grouping may adapt to browser locale in some places, but currency presentation still assumes USD and English-language conventions.

5.5.3 Layout Testing

Test Objective:
To determine whether the layout remains usable if translated strings become longer than the current English text.

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
L7	Review navigation labels, dashboard cards, buttons, and workspace controls at normal desktop width.	Text fits within containers without overlap, clipping, or unreadable wrapping.	[Fill In]	[Fill In]
L8	Review the same pages at narrower widths or responsive breakpoints.	Layout remains stable and text remains readable on smaller screens.	[Fill In]	[Fill In]
L9	Simulate longer label scenarios where possible by reviewing sections likely to expand in translation.	UI components appear capable of wrapping text without major breakage.	[Fill In]	[Fill In]

Observed Risk:
The current layout appears usable in English, but because no actual translation layer exists, full expansion testing has not yet been completed. Long translated strings could create overflow or wrapping problems in headings, buttons, and summary cards.

5.5.4 Directionality Testing

Test Objective:
To determine whether the current interface is prepared for right-to-left localization scenarios.

Test Case	Steps	Expected Result	Actual Outcome	Pass/Fail
L10	Review navigation, buttons, panels, and content alignment for left-to-right assumptions.	Left-to-right layout remains stable for current usage, and RTL assumptions can be identified.	[Fill In]	[Fill In]
L11	Review icon placement, directional arrows, and visual flow in Workspace and Project Details.	Directional elements are identified as potential RTL adaptation points.	[Fill In]	[Fill In]
L12	Evaluate whether current CSS and layout structure appear ready for dir="rtl" support.	Any missing RTL readiness is documented clearly.	[Fill In]	[Fill In]

Observed Risk:
The current system is designed for left-to-right presentation. Directional icons, alignment patterns, and navigation structure are not currently built or tested for RTL use. Therefore, RTL support should be considered unsupported in the current beta version.

5.6 Results and Analysis

Localization testing indicates that Home4U currently demonstrates limited localization readiness rather than full localization support.
Strengths observed in the current implementation include:

- readable and generally consistent English-language UI text
- browser-based formatting support for some date and number output
- layouts that remain visually stable under normal English usage
- responsive page structures that may support some future expansion with refinement

Limitations observed in the current implementation include:

- no multilingual interface support
- no translation or locale resource system
- partial reliance on hardcoded U.S. currency display
- no right-to-left layout support
- no dedicated locale-switching mechanism
- no formal expansion testing with translated strings

Based on these results, Home4U should be described as English-only with limited locale-aware formatting, rather than as a fully localized product.

5.7 Localization Status Summary

Area	Status	Summary
Text	Partial	English text is stable, but no translation system exists.
Formatting	Partial	Some date and number formatting uses browser locale behavior, but currency display is still U.S.-centric.
Layout	Partial	English layouts are stable, but translation expansion has not been fully validated.
Directionality	Issue	RTL support is not currently implemented or tested.

5.8 Planned Improvements

For future versions, the following improvements would be required to achieve stronger localization support:

- externalize all user-facing strings into a translation-ready resource structure
- add locale-aware currency formatting instead of hardcoded dollar presentation
- define a supported locale strategy
- add UI testing for longer translated strings
- evaluate layout updates for right-to-left rendering
- add a language or locale selection mechanism if multilingual support becomes part of the product scope

5.9 Conclusion

The Home4U beta prototype is not yet a fully localized application, but localization testing was still valuable in identifying current strengths and weaknesses. The system performs adequately for English-language use and demonstrates some limited locale-aware formatting behavior, but it does not yet support multilingual content or right-to-left presentation. These findings will help guide future improvements and provide a realistic assessment of localization readiness for Milestone 4.

6. Code Review

6.1 Objective

The purpose of the code review section is to document the team's internal review process, provide evidence that review and feedback occurred during development, and demonstrate that the team participated in external peer review as required by Milestone 4.
This section addresses three goals:

- document how the team reviewed and approved code internally
- provide screenshot-based evidence of internal pull request review activity
- provide evidence that the team reviewed another team's codebase externally

6.2 Internal Code Review Process

The Home4U team used GitHub as the primary platform for organizing code changes, reviewing updates, and maintaining the default branch. Internal code review was intended to support code quality, reduce regressions, and ensure that multiple team members understood important application changes before they became part of the main project baseline.
The team's internal review workflow included the following steps:

1. A team member made changes on a working branch or prepared a feature update.
2. The changes were reviewed by other team members for correctness, usability impact, readability, and consistency with the current milestone scope.
3. Review comments were used to identify missing validation, UI inconsistencies, structural problems, or documentation mismatches.
4. After review feedback was addressed, the changes were merged into the default branch used for grading.

The review process focused especially on:

- frontend route behavior
- backend protected endpoints
- project workflow consistency
- recommendation and analysis behavior
- milestone-document consistency with implemented functionality

Suggested sentence if needed:
Internal code review was used to improve correctness, reduce UI and documentation mismatches, and ensure that multiple team members could understand and defend important implementation decisions.

6.3 Internal Review Priorities

During Milestone 4, the team's internal reviews concentrated on the following areas:

- whether implemented flows matched the intended milestone scope
- whether protected routes and backend actions enforced authentication properly
- whether the saved project workflow behaved consistently across Dashboard, Workspace, and Project Details
- whether recommendation-generation behavior remained stable
- whether major frontend flows remained usable and testable in the deployed beta prototype
- whether deployment, credential, and documentation issues created risk for grading or demonstration

These review priorities were selected because they directly affected beta readiness and milestone submission quality.

6.4 Evidence of Internal Code Review

The final version of this document should include screenshots of actual GitHub pull requests, review comments, or review conversations. These screenshots should show that team members reviewed code rather than only pushing changes directly without inspection.

Suggested items to include:

- screenshot of a pull request overview
- screenshot of inline review comments
- screenshot of follow-up discussion or requested changes
- screenshot of merged review result where appropriate

Suggested table:

Evidence Item	Description	Screenshot Reference
Internal Review 1	Pull request showing feature review for [fill in]	Figure [X]
Internal Review 2	Inline comment or feedback thread for [fill in]	Figure [X]
Internal Review 3	Approval, revision, or merge evidence for [fill in]	Figure [X]

Suggested caption style:

- Figure X. Internal pull request review for [feature or file area].
- Figure X. Example of review comments addressing [issue].
- Figure X. Final merge or review resolution after requested fixes.

6.5 Internal Code Review Findings

This subsection should briefly summarize the kinds of issues the team found during internal review.
Examples of review findings that may be described if they are true:

- UI flow mismatches between wireframes and implementation
- missing access control on protected backend actions
- inconsistent wording across major pages
- incomplete validation or misleading status messaging
- deployment and credentials documentation issues
- gaps between documented behavior and actual implementation

Suggested write-up:
Internal review helped identify several issues affecting beta readiness, including documentation inconsistencies, protected-route and security concerns, and mismatches between claimed behavior and implemented behavior. These reviews were important for improving submission quality before finalizing Milestone 4 deliverables.

6.6 External Code Review

Milestone 4 also requires each team to perform an external review of another team's codebase. The purpose of this review is to demonstrate that the team can evaluate software engineering quality outside its own project and provide useful feedback using the same standards expected for its own work.
The external review should evaluate areas such as:

- repository organization
- coding standards
- branch and pull-request discipline
- deployment clarity
- security practices
- usability and implementation quality
- milestone-document consistency where applicable

Suggested structure:
Team Reviewed: [Fill In]
Repository Reviewed: [Fill In]
Review Date: [Fill In]
Reviewers: [Fill In]

6.7 External Review Summary

Use this subsection to summarize the actual feedback your team gave to another group.
Suggested format:

- what the reviewed team did well
- what risks or bugs were identified
- what documentation or testing gaps were observed
- what security or branch-discipline issues were found
- what actionable recommendations were provided

Suggested paragraph template:
During the external review, the team evaluated another project's repository organization, implementation quality, documentation clarity, and security posture. The review identified strengths in [fill in], while also noting issues in [fill in]. The team provided feedback focused on correctness, maintainability, and milestone readiness rather than only surface-level observations.

6.8 Evidence of External Review

The final version of the milestone document should include screenshots showing that the external review was actually performed.
Suggested evidence:

- screenshot of the reviewed repository or pull request
- screenshot of comments submitted to the other team
- screenshot of written review feedback
- screenshot of any follow-up acknowledgment or discussion if available

Suggested table:

Evidence Item	Description	Screenshot Reference
External Review 1	Reviewed repository or PR	Figure [X]
External Review 2	Submitted feedback comment	Figure [X]
External Review 3	Additional discussion or review note	Figure [X]

6.9 Review Process Assessment

The team's review process improved project maturity by forcing implementation claims, milestone documentation, deployment behavior, and code quality to be evaluated together rather than separately. This was especially important for Milestone 4 because beta readiness depends not only on working code, but also on whether the product can be defended, tested, and reviewed as a coherent engineering deliverable.
Suggested write-up:
The code review process contributed to better project discipline by helping the team identify inconsistencies between implementation, documentation, and deployment behavior. Internal and

external review both supported Milestone 4 goals by emphasizing engineering quality, not just feature completion.

6.10 Limitations

If applicable, disclose any limitations honestly.
Examples:

- not every change used a full PR workflow
- some changes were reviewed informally before later cleanup
- screenshots reflect representative reviews rather than every commit
- branch-protection policy was improved after earlier milestone feedback

Suggested write-up:
One limitation of the team's review process is that not every earlier project change followed the same level of structured PR review. However, by Milestone 4 the team placed greater emphasis on review evidence, security review, and milestone-aligned quality checks in order to improve beta readiness.

6.11 Conclusion

Code review played an important role in improving Home4U during Milestone 4. Internal review helped identify issues in implementation quality, documentation accuracy, and security posture, while external review demonstrated the team's ability to evaluate another project using similar engineering expectations. Together, these review activities supported the transition from a working prototype toward a more credible beta-level software deliverable.
Important note for you before pasting this:

- this section is structurally ready
- but it still needs your real screenshots, reviewed team name, PR links, and actual examples of comments

7. Security Self Check

7.1 Objective

The purpose of the security self check is to identify the most important protected assets in Home4U, document the security controls currently implemented in the beta prototype, and evaluate whether the current system demonstrates acceptable security discipline for Milestone 4.
This review focuses on practical security behaviors currently present in the application, including:

- authentication and access control
- password protection
- input validation
- file upload restrictions
- login abuse protection
- deployment credential handling
- error handling and secure defaults

7.2 Protected Assets

The following assets are considered protected within the current Home4U system:

1. User account records
2. Password hashes
3. JWT authentication tokens
4. Saved room projects
5. Room analysis results and recommendations
6. Uploaded room images
7. Deployment credentials and infrastructure access
8. Backend configuration values such as production secret keys

These assets are important because unauthorized disclosure, modification, or misuse could affect user privacy, application integrity, or cloud deployment safety.

7.3 Password Encryption Strategy

Home4U does not store raw user passwords. Passwords are hashed before storage using bcrypt.

The current backend authentication flow performs the following steps:

- receives the submitted password
- hashes new passwords before storing them in the database
- verifies login attempts by comparing the submitted password against the stored bcrypt hash
- never returns password values to the client

This protects user credentials from direct exposure if the database is accessed improperly.

In the final document, this subsection should include:

- a screenshot of the password hashing code
- a screenshot or example showing that stored passwords appear as hashes rather than plaintext
- a short caption explaining the hashing flow

Suggested write-up sentence:

Home4U stores passwords as bcrypt hashes rather than plaintext values. During authentication, the submitted password is verified against the stored hash, reducing the risk of credential exposure if database records are compromised.

7.4 Authentication and Access Control

Home4U uses JWT-based authentication for protected operations.
Current access control behavior includes:

- unauthenticated users are redirected away from protected frontend routes
- protected backend endpoints require a valid bearer token
- project and recommendation data are restricted to the authenticated owner
- protected project actions verify both authentication and resource ownership

Examples of protected behaviors in the current implementation include:

- viewing room projects
- updating project data
- uploading project photos
- requesting project analysis
- retrieving project recommendations
- marking recommendations complete

Additionally, a previously exposed tag-creation endpoint was revised so that authenticated access is now required for that mutation.
Suggested write-up sentence:
The Home4U backend enforces authentication on protected routes using bearer-token validation and verifies resource ownership for project-specific actions, helping ensure that one user cannot directly access another user's saved project data.

7.5 Login Protection and Abuse Prevention

Home4U includes login rate limiting to reduce the risk of repeated brute-force login attempts.
Current behavior:

- failed logins are tracked
- repeated invalid attempts are limited by both identity and IP-based windows
- the backend returns a blocked response after the configured threshold is exceeded
- successful authentication resets the identity-specific failure count

This improves protection against repeated credential-guessing attempts while preserving usability for valid users.
Suggested write-up sentence:
The authentication system includes rate limiting for repeated failed login attempts, which helps reduce brute-force abuse against user accounts.

7.6 Input Validation

Home4U uses backend validation and request constraints to reduce invalid input and maintain consistent data handling.
Examples of current validation include:

- email validation during signup
- required structured request bodies for project creation and analysis
- constrained numeric ranges for intensity and image-profile fields

- limited accepted values for budget tier and lighting inputs
- search query minimum length and paging limits
- project existence checks and user ownership checks before mutations

Example Validation Areas

Area	Validation Present	Notes
Signup	Yes	Email validation and duplicate-account detection are enforced.
Login	Yes	Credentials are validated and repeated failed attempts are rate-limited.
Search	Yes	Query length and pagination constraints are applied.
Project Analysis	Yes	Structured request schema validates ranges and allowed values.
Protected Resource Access	Yes	Project and recommendation ownership checks are enforced.

Suggested write-up sentence:
Home4U validates key user inputs at the backend layer through schema validation, constrained request parameters, and resource-ownership checks before performing protected actions.

7.7 Search Input Validation

Search was specifically included in the milestone instructions as an example area for input validation review.
Current search protections include:

- a required query parameter
- minimum query length enforcement
- bounded limit values
- bounded page values
- server-side handling that avoids unrestricted free-form execution behavior

The search implementation performs fuzzy style matching against stored style and tag data, but it does so inside the application logic rather than by executing arbitrary user-supplied commands or unsafe dynamic behavior.
Suggested write-up sentence:
The search feature validates required query input and constrains paging parameters, helping ensure that user search requests remain predictable and safe for backend processing.

7.8 File Upload Security

Home4U allows authenticated users to upload room images to projects. This creates additional security risk, so the upload path includes several restrictions.
Current upload protections include:

- authentication required before upload
- project ownership verification
- allowed content types limited to JPEG, PNG, and WebP
- file size restriction enforced
- server-generated stored filenames rather than trusting client filenames

These controls reduce the risk of unauthorized uploads, oversized files, and unsafe filename behavior.

Upload Validation Summary

Check	Implemented	Description
Authentication required	Yes	Uploads are restricted to authenticated users.
Ownership check	Yes	Users may upload only to their own projects.
File type restriction	Yes	Only JPG, PNG, and WebP are accepted.
File size restriction	Yes	Oversized files are rejected.
Server-side filename generation	Yes	Stored filenames are generated by the backend.

7.9 Error Handling and Secure Response Behavior

The backend includes centralized exception handling and applies baseline security-related response headers.
Current security-related response behavior includes:

- generic internal server error responses instead of exposing full stack traces to clients
- response headers such as X-Content-Type-Options
- frame and referrer protection defaults
- permission-policy restrictions for unused browser capabilities

This helps reduce unnecessary information exposure and provides a safer default response posture.
Suggested write-up sentence:
Home4U uses centralized backend error handling and baseline security headers to reduce accidental information leakage and strengthen default response behavior.

7.10 Deployment Credential Handling

Infrastructure access was previously a major security concern because a live SSH private key had been committed into the repository. That key has since been rotated and retired from active server access.

The team’s corrected security posture is:

- the old leaked SSH key is no longer accepted by the EC2 server
- a rotated replacement key is used outside the repository
- the repository no longer serves as a distribution point for live private keys
- credential documentation now instructs users to obtain access through an approved secure channel

This issue should still be disclosed honestly in the report because it was a real security risk discovered during Milestone 4 preparation.

Suggested write-up sentence:

During Milestone 4 preparation, the team identified that a server access key had been improperly committed to the repository. The key was rotated, the old key was retired from the server, and repository guidance was updated so that live private keys are no longer distributed through version control.

7.11 Production Secret Management

The backend now requires HOME4U_SECRET_KEY in production rather than silently relying on the development fallback secret.

This is important because:

- JWT signing should not depend on a known default secret in a public deployment
- production startup should fail safely if a required secret is missing
- deployment documentation should make the production secret requirement explicit

Suggested write-up sentence:

The production backend now requires an explicit application secret key for JWT signing, preventing deployment with an insecure default secret.

7.12 Security Findings Summary

Strengths

- Passwords are hashed with bcrypt
- Protected routes require authentication
- Project ownership checks are enforced
- Login abuse protection is implemented through rate limiting
- Upload restrictions are enforced
- Production secret-key requirements are strengthened
- Auth is now required for protected tag mutation behavior
- Leaked SSH key was rotated and retired from server access

Remaining Risks / Limitations

- The current deployment still uses HTTP rather than a trusted HTTPS configuration
- Full secret-management maturity is still limited compared with production-grade enterprise standards
- Localization and accessibility reviews are separate from security and do not replace deeper penetration testing
- Broader security testing such as dependency auditing, automated vulnerability scanning, and formal penetration testing has not yet been fully documented in this milestone

7.13 Conclusion

The Home4U beta prototype demonstrates meaningful security improvements and includes several practical protections appropriate to its current scope, including password hashing, authenticated access control, ownership checks, login rate limiting, upload validation, and safer production secret handling. The most significant security issue identified during Milestone 4 preparation was the exposure of an active SSH private key in the repository. That issue was corrected through key rotation, retirement of the old key, and updated credential-handling guidance.

Overall, the system shows reasonable beta-level security discipline, but the team should continue improving deployment hardening, transport security, and documented operational security practices in future revisions.

8. Non-Functional Requirements Status

8.1 Objective

This section evaluates the current status of the non-functional requirements originally defined in Milestone 1 Version 2. Based on the Milestone 1 Version 2 document in the repository, the non-functional requirement categories used by the team were:

- Performance
- Reliability
- Security
- Usability
- Scalability
- Maintainability

Each category is evaluated here using the status labels DONE, ON TRACK, or ISSUE based on the current Home4U beta prototype, the deployed application state, and the engineering evidence available in the repository during Milestone 4 preparation.

8.2 Status Definitions

DONE: The requirement is substantially satisfied by the current system and supported by implementation evidence.
ON TRACK: The requirement is partially satisfied or progressing appropriately, but still needs refinement, broader testing, or stronger evidence.
ISSUE: The requirement is not yet adequately satisfied or still contains important limitations that affect milestone readiness.

8.3 Non-Functional Requirements Status Table

Category	Status	Justification
Performance	ON TRACK	The system currently supports normal beta-level interaction across Dashboard, Workspace, and Project Details, and the frontend build, linting, and smoke-test flows pass successfully. However, formal performance benchmarking, response-time reporting, and load-oriented validation have not yet been fully documented in the milestone evidence.
Reliability	ON TRACK	The current implementation includes centralized backend error handling, project existence checks, protected resource ownership checks, upload validation, and stable smoke-test coverage for major backend flows. The application demonstrates stable normal-use behavior, but reliability evidence is still limited to beta-level testing rather than deeper fault-tolerance or recovery analysis.
Security	ON TRACK	The system includes bcrypt password hashing, JWT-based authentication, protected-route enforcement, login rate limiting, upload restrictions, and ownership checks on project resources. During Milestone 4 preparation, a serious infrastructure key-exposure issue was discovered and corrected through SSH key rotation and retirement of the old key. Security has improved significantly, but the deployment still lacks full HTTPS hardening and broader production-grade security maturity.

Usability	ON TRACK	The system includes major user flows for style selection, workspace launch, room analysis, saved project review, and recommendation tracking. Frontend route and accessibility smoke tests pass, and major pages are navigable. However, the team still needs complete usability-test evidence, participant results, and full closure of earlier wireframe-fidelity concerns before this area can be considered fully complete.
Scalability	ISSUE	The current deployment architecture is still limited to a single application instance with a SQLite-backed data layer and does not yet demonstrate true scaling support beyond the class project scope. While the backend and frontend are structured in a maintainable way, the present infrastructure does not provide strong evidence of horizontal scalability or production-scale growth handling.
Maintainability	ON TRACK	The repository has a clear application structure, separated frontend and backend concerns, reusable components, organized route modules, and automated smoke-test coverage for key flows. Documentation quality has improved, and recent fixes aligned the repository more closely with safer operational practices. However, maintainability is still affected by the need for stronger evidence of review discipline, milestone-document consistency, and continued cleanup of remaining process gaps.

8.4 Detailed Status Discussion

8.4.1 Performance

Home4U currently performs adequately for a beta prototype under normal user interaction. The application can load major pages, perform authenticated routing, execute style search behavior, and generate room-analysis results without major instability during standard testing. Frontend validation confirms that the production build succeeds and that smoke tests pass for key route and accessibility flows.

Despite this, the project does not yet include fully documented timing benchmarks, stress measurements, or formal response-time reporting across multiple environments. Because of that, performance is best classified as ON TRACK rather than fully complete.

8.4.2 Reliability

Reliability in the current beta prototype is supported by several practical safeguards. The backend handles missing projects, missing analysis state, invalid recommendation conditions, and unhandled exceptions with structured responses rather than full application crashes. Project analysis, login rate limiting, and search flows also have passing smoke-test coverage in the local project environment.

Even with these strengths, the current evidence is still centered on ordinary beta testing rather than fault injection, failover design, or advanced recovery testing. For this reason, reliability is currently ON TRACK.

8.4.3 Security

Security is one of the strongest improved areas in the current Milestone 4 state. Passwords are stored as bcrypt hashes, authenticated actions require bearer-token validation, user-owned project data is access-controlled, repeated invalid logins are rate-limited, and upload behavior is validated for file type and size. In addition, the production configuration now requires a non-default secret key rather than silently accepting an insecure fallback.

A significant issue was identified during Milestone 4 preparation: an active SSH private key had been committed to the repository. That key was rotated, replaced, and retired from active server access. This was an important correction, but because transport security and broader production-grade hardening are still incomplete, security remains ON TRACK rather than fully done.

8.4.4 Usability

Usability is supported by the existence of a coherent user flow through style browsing, workspace setup, recommendation generation, and saved project review. Protected routing behaves consistently, accessibility-focused route tests pass, and the interface supports major beta-level room-planning tasks.

However, formal usability evidence still depends on real participant testing, effectiveness and efficiency measurements, and final user satisfaction results. In addition, previous milestone feedback identified wireframe-fidelity issues that still need complete documentation closure. Therefore, usability is ON TRACK.

8.4.5 Scalability

Scalability remains the weakest non-functional area at the current stage. The system is structured like a full-stack web application, but the deployed beta environment still uses a single EC2-hosted instance and a SQLite-backed data layer. This architecture is sufficient for milestone demonstration and basic beta use, but it does not provide strong evidence of scaling under larger production demand.

Because true scalability support is not yet demonstrated in implementation or testing evidence, this category is best classified as an ISSUE.

8.4.6 Maintainability

Maintainability is supported by the clear directory structure, separation of concerns between backend and frontend, reusable UI components, defined API route modules, and the presence of smoke-test coverage for important flows. The repository also shows continued effort to align documentation and code behavior more closely during milestone preparation.

At the same time, maintainability still depends on continued consistency in reviews, documentation updates, and milestone alignment. Since the structure is solid but process maturity is still improving, maintainability is classified as ON TRACK.

8.5 Summary

Overall, the Home4U beta prototype shows meaningful progress across most non-functional categories. Performance, reliability, security, usability, and maintainability all demonstrate substantial implementation progress and are currently ON TRACK, while scalability remains an ISSUE because the deployed architecture does not yet show strong growth-readiness beyond the current class project scope.

This status review reflects the current beta state honestly: the system is increasingly stable and defensible for Milestone 4, but several non-functional areas still require stronger evidence and refinement before final delivery

9. Team Contributions

Team Member	Role(s)	Contribution Summary	Score (1-10)
Caleb Ponce	Team Lead / System Architecture	Coordinated milestone progress, managed repository and deployment readiness, reviewed system-wide issues affecting milestone delivery, supported backend and documentation alignment, and helped ensure that the beta prototype, security fixes, and milestone requirements remained consistent.	9

Mason Lee	Data Modeling & Scoring Engine	Contributed to data modeling, scoring logic, and project-analysis behavior, and supported the recommendation structure used in the room-style evaluation workflow. Also contributed to milestone-related repository updates and supporting implementation work.	8
Christopher Quach	Frontend Development	Contributed to frontend implementation across the major user-facing pages, including interface behavior, navigation flow, and visual presentation improvements that support the Dashboard, Workspace, Project Details, and related user workflows.	8
Tyler Morris	Backend & AI Integration	Contributed to backend implementation and analysis-related behavior, including recommendation-related logic and workflow support for the room-analysis pipeline. Also supported milestone implementation areas tied to the project's AI-labeled or algorithmic recommendation features.	9
Dias Almat	Technical Writer	Contributed to milestone documentation, document organization, written deliverables, and formatting support. Helped maintain milestone content structure and supported the written presentation of the project's requirements, scope, and submission materials.	9